GROUP ASSIGNMENT

Network Thinking and Agent Based Modeling

Atanasova Teodora
Hoxha Enio

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RESEARCH QUESTION

1 HOW DO **DRUG TRAFFICKING NETWORKS ADAPT** TO THE OCCUPATION OF KEY HUBS AND **INCREASED ENFORCEMENT** PRESSURE?

2 WHICH **ENFORCEMENT STRATEGIES ARE MOST EFFECTIVE** TO DISRUPT DRUG TRAFFICKING NETWORKS?

EXPECTATIONS

Based on existing literature and personal expectations

1. INEFFICIENCY of RANDOM ENFORCEMENTS ✓
2. EFFICIENCY OF TARGETING MAIN HUBs ✓
3. RESILIENCE of THE NETWORK ✓
4. COLLAPSE OF THE NETWORK X

LITERATURE

1. BALLOON EFFECT (Dorn et al. 2005): A hydraulic model where localized interdiction shifts production or trafficking geographies rather than eliminating them

2. SCALE-FREE NETWORKS (Barabási et al. 1999): A topology where a few hubs concentrate connectivity, serving as critical yet resilient bottlenecks.

3. INEFFECTIVENESS OF CRIMINAL NETWORK DISTRUPTION (Duijn et al. 2014) counterintuitive dynamic explains targeted disruption of criminal networks can paradoxically increase their cohesion and resilience

4. MADTOR Project (Manzi, D., & Calderoni, F. 2024): Network Model for Assessing Drug Trafficking Organizations Resilience under different law enforcement pressures.

LITERATURE

What our model adds compared to existing ones

Feature / Dimension	NarcoLogic (Magliocca et al.)	MADTOR (Manzi & Calderoni)	Duijn / Morselli (Resilience Lit.)	OUR MODEL
Primary Scale	Geographic/Spatial	Case-Specific (Italy)	Static/Structural	**Topological/Systemic**
Agent Behavior	Spatial Displacement	Fixed Response	Theoretical Trade-off	Organizational Culture (Adaptive vs. Stubborn)
Enforcement	Spatial Intervention	Individual Arrests	Node Removal	Factorial Design (Targeting, Frequency, Duration)
Adaptation	Movement of nodes	Fixed thresholds	Instantaneous change	Endogenous Growth (Tick-by-tick route building)
Key Metric	Geographic shift	Connectivity loss	Qualitative Paradox	Flow Displacement Index (FDI) & Clustering Rate %, Seizure Rate

THE MODEL

Introduction

WHAT IT IS ABOUT

The model showcases how intelligent versus random interdiction strategies interact with internal cartel risk tolerance to drive spatial displacement.

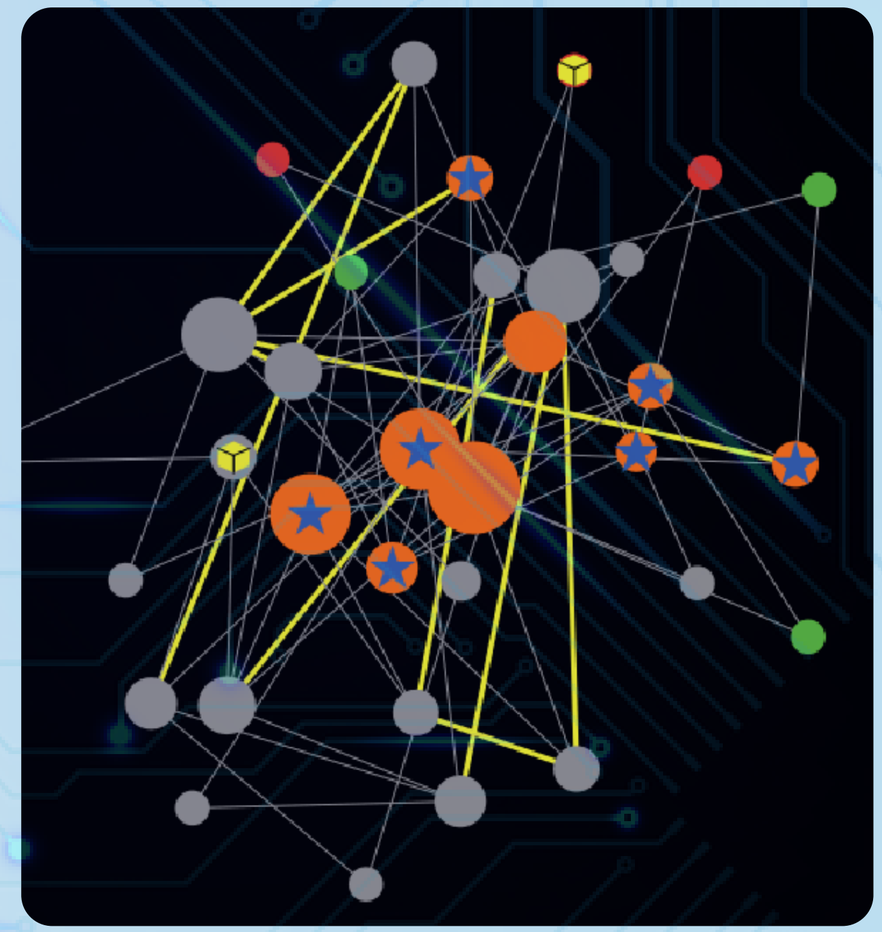
WHAT IT MONITORS

Structural Weaknesses: Identify how targeted hub occupation impacts the overall throughput and connectivity of scale-free networks

Dark Resilience: Evaluate the transition from "parasitic" reliance on public infrastructure to "independent" clandestine corridors.

THE MODEL

Components

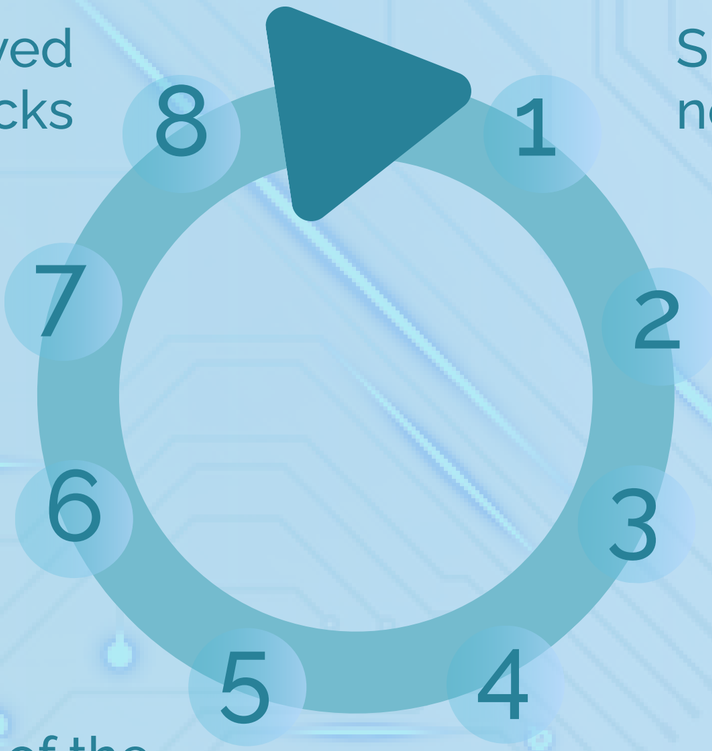


- supply sources spawning packages
- market demand endpoint
- transit nodes
- perceived police interdiction zone
- original transit routes
- new emergent routes for adaptation
- active presence of enforcement
- drug package

THE MODEL

Core loop

1. Spawn of packages on (Green nodes)
2. Shipments utilize the cheapest public infrastructure (Gray Hubs).
3. Intelligence led police raid the busiest hub
4. Node risk instantly spikes to 100. Shipments currently in the hub seized
5. Immediate spatial displacement of the packages on clandestine routes (Yellow links)
6. Packages reach destination (Red nodes)
7. RIsk decays over time and traffick returns
8. Unused links are removed after 180 ticks



BEHAVIOURS

Law enforcement

WHERE DOES IT TARGET?

Based on targeting intelligence: if “Blind Patrols” then randomly,
if “Intelligence-Led Patrols” then on the busiest transit node

HOW OFTEN DOES IT TARGET?

Based on enforcement chance slider. Represents a proxy for state resources

HOW LONG DOES IT TARGET?

Based on enforcement duration slider. Represents a proxy for state resources

PROXY INTERPRETATION

The duration and frequency of target could be a representation of the available state resources. The intelligence represents amount of information available



BEHAVIOURS

Traffickers

HOW DOES IT CHOOSE WHERE TO GO?

ROUTING MATH

$$W_{\text{link}} = C_{\text{base}} + R_{\text{node1}} + R_{\text{node2}}$$

Traffickers minimize "Route Weight" using bounded rationality.

MEMORY SYSTEMS

- When police raids a Node, its risk spikes to 100
- Risk decay: perceived risk drops by 5 points per tick after enforcement leaves, reaching zero in 20 ticks.

PROXY INTERPRETATION

The meaning of the links is not strictly geographical. It represents also bribes and collaborations between criminals



BEHAVIOURS

Traffickers

WHEN DO THEY CREATE AND REMOVE NEW ROUTES?

CREATION OF NEW CLANDESTINE ROUTES

When the %Seized of last 90 days surpasses the “loss tolerance threshold” of the Cartel’s. two behaviours: “Stubborn” vs “Adaptive”. They do so by connecting the safe neighbors nodes to bypass the orange hubs

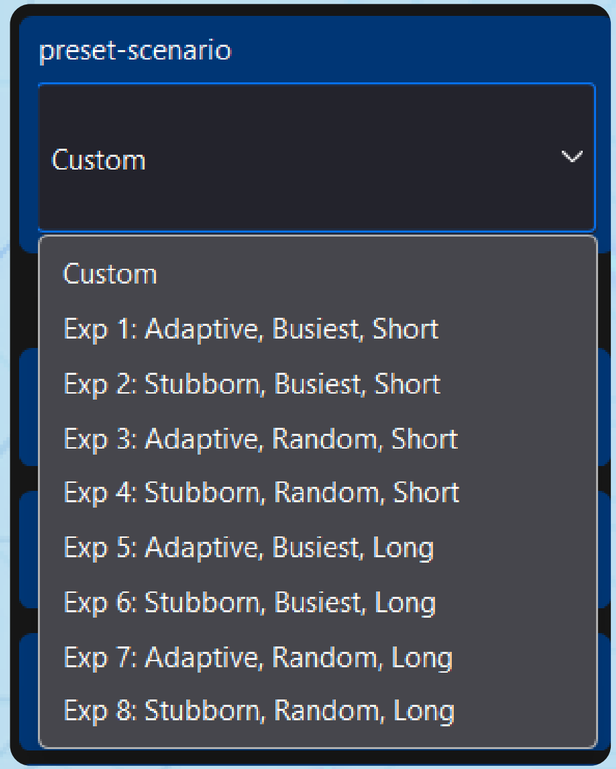
REMOVAL OF ROUTES

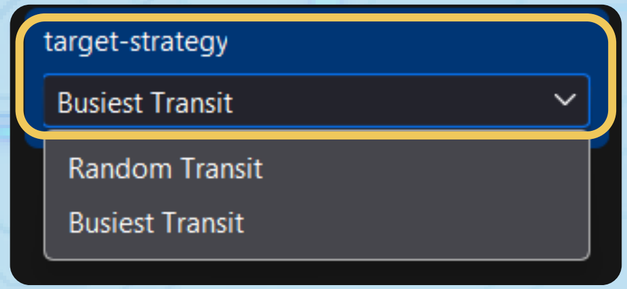
After 180 days the routes that have not been used are removed

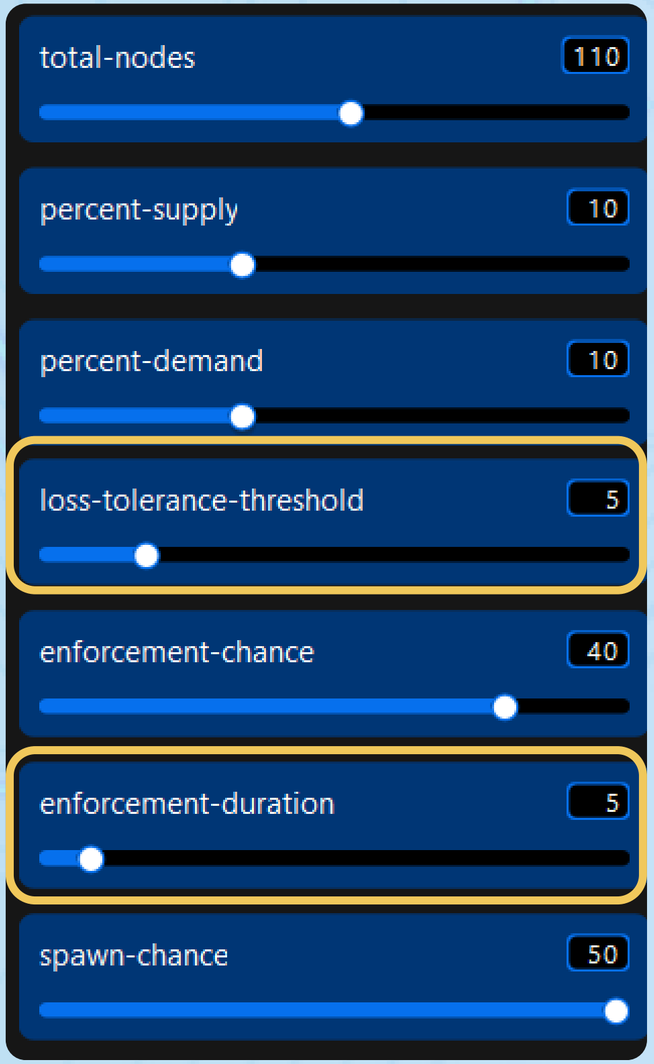
PROXY INTERPRETATION

The behaviour of the cartel is not only interpretable as strictly loss tolerant or averse, but also represents their ability, experience and resources to create new links

STRATEGY & SLIDERS

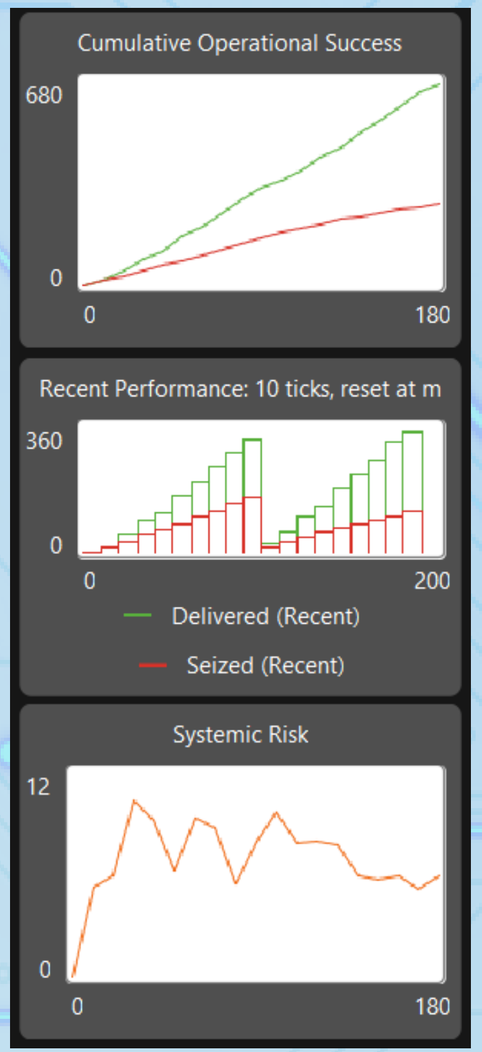






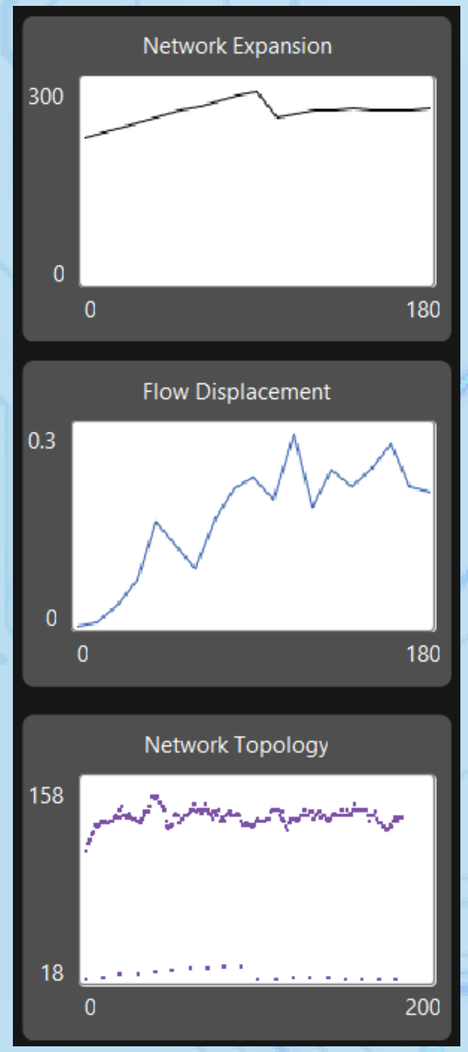
Only the circled parameters vary between experiments, allowing clean comparability between scenarios

PLOTS AND MONITORS



in the seized fall both packets that are intercepted and packets that can not find any path at all

tracks the network fragility. If network relies too heavily on few hubs, then it spikes



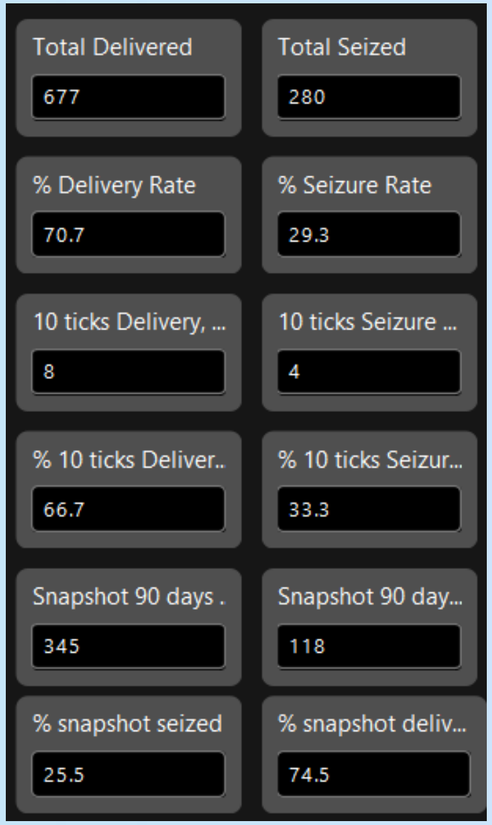
represents total link count over time

traffic on yellow links/total traffic

FDI = 0 means all traffic uses gray routes

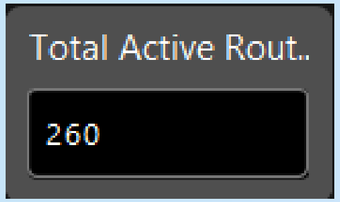
clustering coefficient.

High Clustering: Nodes = form "cliques" or tight groups. If one path is blocked, many local alternatives exist.



TOTAL CUMULATIVE

90 DAYS RESET



LIVE DEMO

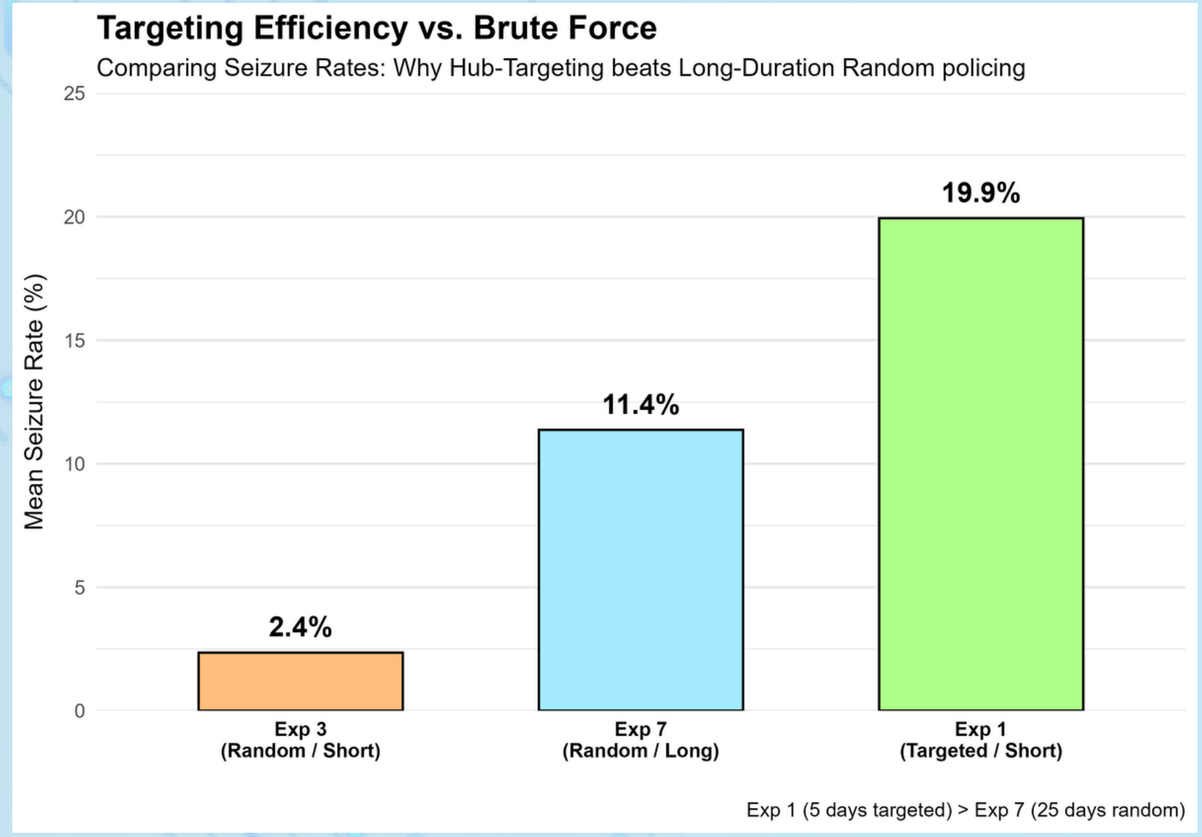
THE 8 EXPERIMENTS

Scenarios allow controlled comparability. Data on the scenarios have been collected with statistical robustness, running 30 times each experiment

Experiment	Cartel type	Loss Tolerance	Target Strategy	Enforcement Duration	Description of the Experiment
Exp 1	Adaptive	5%	Busiest	Short — 5	Balloon Effect baseline. Reactive cartel under intelligent, brief enforcement.
Exp 2	Stubborn	30%	Busiest	Short — 5	Rigid cartel absorbs losses under intelligent enforcement before adapting.
Exp 3	Adaptive	5%	Random	Short — 5	Reactive cartel under blind enforcement. Hubs rarely targeted.
Exp 4	Stubborn	30%	Random	Short — 5	Rigid cartel, blind enforcement. Minimal pressure from both sides.
Exp 5	Adaptive	5%	Busiest	Long — 25	Sustained occupation forces permanent shadow-route investment.
Exp 6	Stubborn	30%	Busiest	Long — 25	Can sustained targeted pressure break even a rigid cartel?
Exp 7	Adaptive	5%	Random	Long — 25	Long blind raids on reactive cartel. Duration without intelligence.
Exp 8	Stubborn	30%	Random	Long — 25	Worst-case enforcement: long, blind, against a rigid cartel.

DISCUSSION OF RESULTS

R1: RANDOM STRATEGY IS INEFFECTIVE

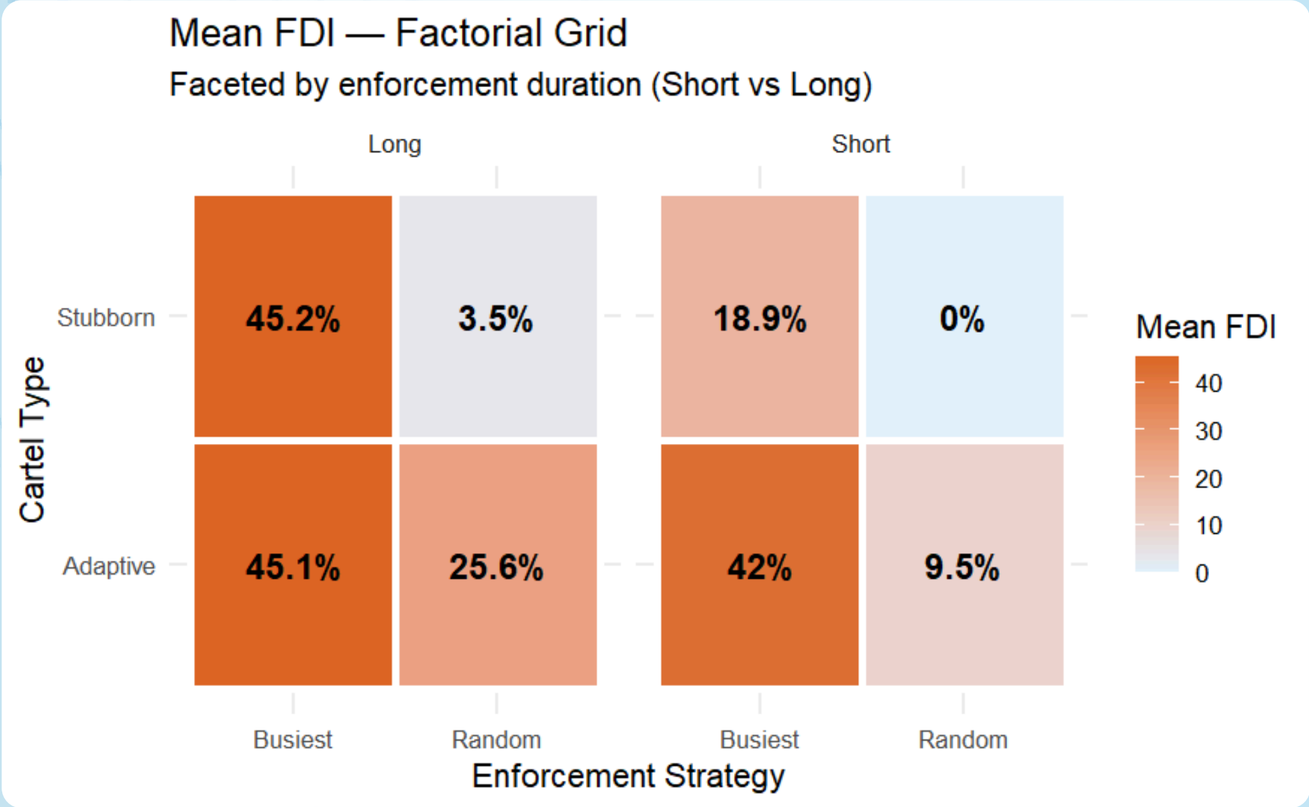


TAKEAWAY:

Holding a random node for 25 days (Exp 7) is half as effective as holding the right node for 5 days (Exp 1).

DISCUSSION OF RESULTS

R2: BALLOON EFFECT INFLUENCED BY CARTEL CULTURE



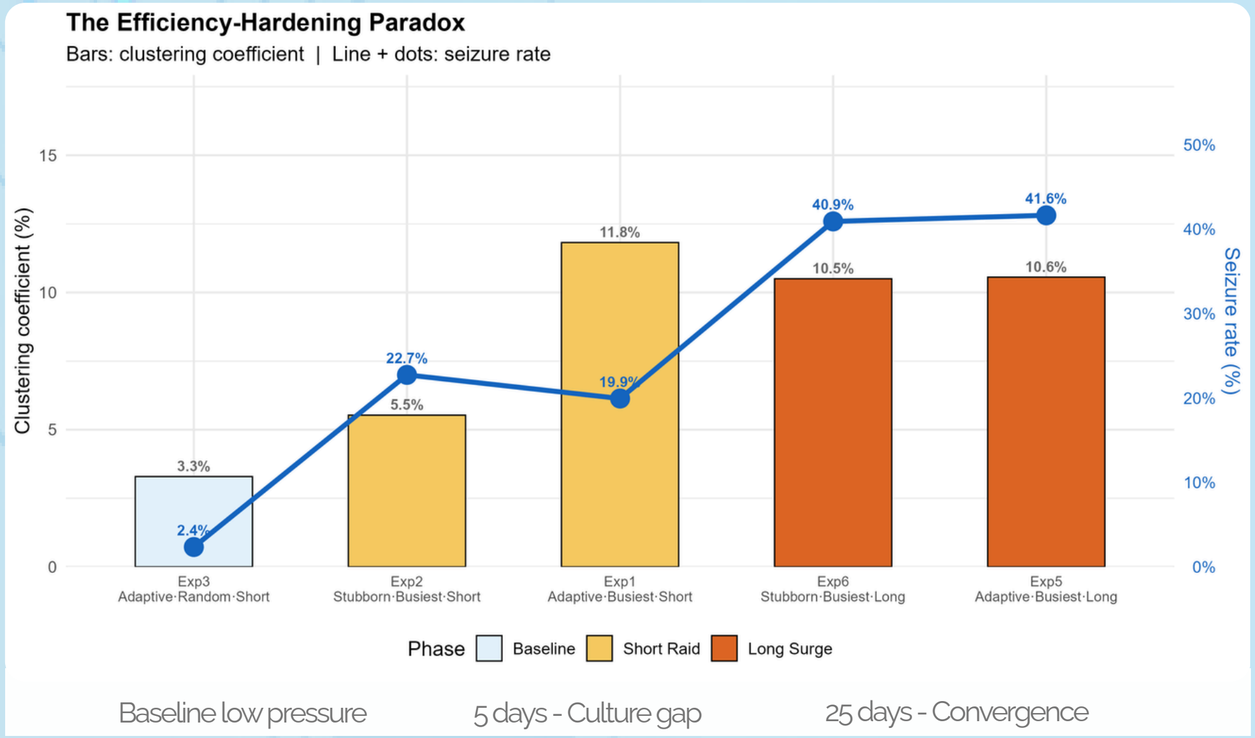
TAKEAWAY:

Adaptive cartel types adapt the most regardless of the enforcement strategy.
Stubborn cartel types reacy only under the highest pressure.

NB: The cartel being “adaptive” or “stubborn” does not only represent their loss aversion but also represents their ability, experience and resources to create new links.
ex: “adaptive” spends more money for bypass routes

DISCUSSION OF RESULTS

R3: EFFICIENT ENFORCEMENT CAUSES HIGH RESILIENCE



TAKEAWAY:

Targeted enforcement acts as an "evolutionary pressure" that forces structural hardening. This suggests that sustained targeted pressure creates a "Structural Destiny."

! **PARADOX** (Duijn 2024) ! :
successful policing is the cause of network resilience, and makes it harder to break, harder to map, and functionally displaced

DISCUSSION OF RESULTS

R4: DO CARTELS EVER DIE?

TAKEAWAY:

Running all the experiments for a very long time, has shown us that cartels **never die.**
However this result is highly dependent on the preset values.

DISCUSSION OF RESULTS

Summarising table

Type	Experiment	Target	Duration	Runs	Avg Delivered	Avg Seized	Seizure Rate	FDI (Flow %)	Clustering
Adaptive	Exp1_Adaptive_Busiest_Short	Busiest	Short	30	4,392.53	1,093.70	19.94	42.03	11.82
	Exp3_Adaptive_Random_Short	Random	Short	30	5,345.87	128.73	2.35	9.53	3.29
	Exp5_Adaptive_Busiest_Long	Busiest	Long	30	3,212.23	2,293.53	41.64	45.15	10.56
	Exp7_Adaptive_Random_Long	Random	Long	30	4,867.53	624.90	11.37	25.59	4.98
Stubborn	Exp2_Stubborn_Busiest_Short	Busiest	Short	30	4,226.47	1,243.47	22.74	18.94	5.53
	Exp4_Stubborn_Random_Short	Random	Short	30	5,366.47	122.90	2.24	0.00	1.54
	Exp6_Stubborn_Busiest_Long	Busiest	Long	30	3,240.50	2,243.83	40.92	45.22	10.50
	Exp8_Stubborn_Random_Long	Random	Long	30	4,901.13	567.43	10.38	3.55	1.94

CONCLUSION & IMPLICATIONS

1 INTELLIGENCE > RESOURCES

Targeted raids (5 days) > Random patrols (25 days) with seizure rate of 19.9% vs. 11.4%, using 1/5th of the resources

POLICY IMPLICATION: Budget should be reallocated from "Operational Volume" (patrol hours) to "Targeting Quality" (intelligence).

2 SUCCESS PARADOX

Experiments with highest seizures (~41%) => highest Network Hardening (10.5% Clustering) and FDI (45%).
In a way it is training the network to be more resilient, decentralized and harder to map for the next. Even the stubborn type of markets adapted in the end.

POLICY IMPLICATION: Seizure volume is a misleading metric. Agencies optimizing for seizures are unintentionally incentivizing cartels to build permanent, clandestine "shadow infrastructure" that wouldn't exist otherwise.

3 NEVER ZERO FLOW

Even under maximum sustained pressure, 58% of flow still reached the destination.

POLICY IMPLICATION: Computational evidence suggests "elimination" is an impossible benchmark. Policy success should be reframed as containment and cost-raising rather than "victory" or "defeat."

LIMITATIONS & FUTURE EXPANSIONS

LIMITATIONS

1. Static Intelligence: Agents possess perfect knowledge or total ignorance.
2. Latent Geography: Lack of physical terrain constraints.
3. No economic constraints

FUTURE EXPANSIONs

1. Dynamic learning algorithms.
2. Calibration with real-world narcotrafficking data.
3. Variations in value and risk profile

THANK YOU FOR LISTENING.

WE ARE HERE FOR YOUR QUESTIONS

Atanasova Teodora
Hoxha Enio

le Fin. 🐢